Foundations of Statistical Learning & Algorithms

Today's goal • 3/3 items

Module 1
Introduction to Linear Algebra and Vector Spaces

Getting Started

- Course Introduction
  Video • 1 min
- Course Overview
  Reading • 2 min
- Syllabus
  Reading • 10 min
- Meet Your Instructor
  Video • 1 min
- Meet Your Fellow Learners
  Discussion Prompt • 10 min
- Academic Integrity
  Reading • 1 min

Lesson 1: Introduction to Linear Algebra

- Introduction to Machine Learning
  Reading • 25 min
- Introduction to Linear Algebra
  Reading • 4 min
- Why Linear Algebra and Mathematics?
  Reading • 2 min
- Notation
  Reading • 5 min

Lesson 2: Systems of Linear Equations

- Foundational Concepts of Systems of Linear Equations
  Reading • 20 min
- [H5P] System of Linear Equations
  Ungraded App Item • 20 min
- Solved Example and Recommended Resources
  Reading • 20 min
- Check Your Knowledge: System of Linear Equations
  Graded Assignment • Grade: 100%

Lesson 3: Matrices

- Matrices and Matrix Operations
  Reading • 20 min
- Matrices
  Video • 8 min
- Helpful Resources and Solved Examples
  Reading • 10 min
- Check Your Knowledge: Matrices
  Graded Assignment • 16 min

Lesson 4: Vector Spaces

- Foundations of Vector Spaces: Operations and Subspaces
  Reading • 20 min
- Vector Space
  Video • 7 min
- Vector Space Properties
  Reading • 10 min
- Subspace
  Reading • 1 min
- Check Your Knowledge: Vector Spaces
  Graded Assignment • Grade: 85.71%

Lesson 5: Orthogonal Complement and Projection

- Introduction to Orthogonal Complement
  Reading • 20 min
- Introduction to Orthogonal Projections
  Reading • 40 min
- **Importance of Projections**
  Reading • 4 min
- Projection Onto One-Dimensional Subspaces (Lines)
  Reading • 15 min
- Projection Onto General Subspaces
  Reading • 10 min
- Projections Case Study
  Reading • 10 min

Module 2
Linear Transformation and Matrix Decomposition

Module 3
Probability Foundations for Statistical Learning

Module 4
Introduction to Model Evaluation and Optimization

Importance of Projections

0:02 / 1:02 2x

Definition and Importance of Projections:

- A projection is a linear transformation that maps vectors onto a subspace.
- Orthogonal projections minimize the distance between the original vector and its projection, retaining essential data features while reducing dimensionality.
- The projection matrix P associated with a subspace projects any vector onto that subspace, preserving orthogonality.

This image illustrates the concept of **orthogonal projection** of points onto a line in a 2D space. The **black line** represents a subspace (a $1D$ line through the origin), while the **blue dots** are vectors (points) in the space. Each point is projected orthogonally onto the line, with the **orange lines** representing the perpendicular distance between the original points and their projections. The orange points on the line are the orthogonal projections, which are the closest points on the subspace to the original vectors, satisfying the condition that the difference (residual) is orthogonal to the subspace.

Go to next item ✓ Completed

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